Your answers are saved automatically.

Remaining Time: 03 minutes, 51 seconds.

Question Completion Status:

1 2 3 4 5 6 7 8 9 10 11 12 13 14
27 28 29 30 31 32 33 34 35 36 37 38 39 40

QUESTION 1

Consider a **0/1 knapsack problem** with three items:
Item 1: weight = 1 and benefit = 20
Item 2: weight = 2 and benefit = 30
Item 3: weight = 3 and benefit = 40
The knapsack capacity = 4.
V[1,1] =

- 0
- 20
- 30
- 50

QUESTION 2

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Remaining Time: 03 minutes, 47 seconds.

Question Completion Status:

1 2 3 4 5 6 7 8 9 10 11 12 13
27 28 29 30 31 32 33 34 35 36 37 38 39

QUESTION 2

Consider a **0/1 knapsack problem** with three items:
Item 1: weight = 1 and benefit = 20
Item 2: weight = 2 and benefit = 30
Item 3: weight = 3 and benefit = 40
The knapsack capacity = 4.
V[2,2] =

- 20
- 30
- 50
- 60

QUESTION 3

Consider a **0/1 knapsack problem** with three items:
Item 1: weight = 1 and benefit = 20

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Remaining Time: 03 minutes, 43 seconds.

Question Completion Status:

1 2 3 4 5 6 7 8 9 10 11 12 13
27 28 29 30 31 32 33 34 35 36 37 38 39

- (●) 30
- ( ) 50
- ( ) 60

QUESTION 3

Consider a **0/1 knapsack problem** with three items:
Item 1: weight = 1 and benefit = 20
Item 2: weight = 2 and benefit = 30
Item 3: weight = 3 and benefit = 40
The knapsack capacity = 4.
Which items are included in the knapsack?

- ( ) Only item three
- ( ) Items 1 and 2
- (●) Items 1 and 3
- ( ) All the three items are included

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Remaining Time: 03 minutes, 38 seconds.

Question Completion Status:

1 2 3 4 5 6 7 8 9 10 11 12 13
27 28 29 30 31 32 33 34 35 36 37 38 39

- Only item three
- Items 1 and 2
- Items 1 and 3
- All the three items are included

QUESTION 4

Consider a **0/1 knapsack problem** with three items:
Item 1: weight = 1 and benefit = 20
Item 2: weight = 2 and benefit = 30
Item 3: weight = 3 and benefit = 40
The knapsack capacity = 4.

V[0,0] =

- 0
- 20
- 30
- 50

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Remaining Time: 03 minutes, 34 seconds.

Question Completion Status:

1 2 3 4 5 6 7 8 9 10 11 12
27 28 29 30 31 32 33 34 35 36 37 38

30

50

QUESTION 5

Consider a **0/1 knapsack problem** with three items:
Item 1: weight = 1 and benefit = 20
Item 2: weight = 2 and benefit = 30
Item 3: weight = 3 and benefit = 40
The knapsack capacity = 4.
V[3,3] =

20

30

50

60

QUESTION 6

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Remaining Time: 03 minutes, 30 seconds.

Question Completion Status:

- (●) 50
- ( ) 60

QUESTION 6

Consider a **0/1 knapsack problem** with three items:
Item 1: weight = 1 and benefit = 20
Item 2: weight = 2 and benefit = 30
Item 3: weight = 3 and benefit = 40
The knapsack capacity = 4.
The maximum benefit of items included in the knapsack is stored in:

- ( ) V[0,0]
- (●) V[3, 4]
- ( ) V[3,3]
- ( ) V[2,4]

QUESTION 7

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Remaining Time: 03 minutes, 27 seconds.

Question Completion Status:

1 2 3 4 5 6 7 8 9 10 11 12
27 28 29 30 31 32 33 34 35 36 37 38

- V[3, 4]
- V[3,3]
- V[2,4]

QUESTION 7

Consider a **0/1 knapsack problem** with three items:
Item 1: weight = 1 and benefit = 20
Item 2: weight = 2 and benefit = 30
Item 3: weight = 3 and benefit = 40
The knapsack capacity = 4.
What is the maximum benefit of items included in the knapsack?

- 20
- 30
- 50
- 60

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Remaining Time: 03 minutes, 22 seconds.

Question Completion Status:

1	2	3	4	5	6	7	8	9	10	11
27	28	29	30	31	32	33	34	35	36	37

QUESTION 8

Consider a **0/1 knapsack problem** with three items:
Item 1: weight = 1 and benefit = 20
Item 2: weight = 2 and benefit = 30
Item 3: weight = 3 and benefit = 40
The knapsack capacity = 4.
V[2,3] =

- 20
- 30
- 50
- 60

QUESTION 9

Consider the recurrence: $T(n) = T(n/2) + n^2$, using the Master method, choose

Click Save and Submit to save and submit. Click Save All Answers to save all answe

QUESTION 9

Consider the recurrence: $T(n) = T(n/2) + n^2$, using the Master method, choose the correct answer.

- Apply case 1 and $T(n) = \Theta(1)$
- Apply case 2 and $T(n) = \Theta(\log n)$
- **Apply case 3 and $T(n) = \Theta(n^2)$**
- Master method cannot be used.

QUESTION 10

Consider the recurrence: $T(n) = T(n/2) + 2n$, the size of each subproblem produced at each step is:

- equal to the size of the original problem
- **half the size of the original problem**
- double the size of the original problem
- All the above answers are incorrect

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

QUESTION 11

Consider the recurrence: $T(n) = T(n/2) + 3n$, the number of subproblems produced at each step is:

- (●) 1
- ( ) 2
- ( ) 3
- ( ) Cannot be determined

QUESTION 12

Consider the recurrence: $T(n) = 2\,T(n/2) + n$, using the Master method, choose the correct answer.

- ( ) Apply case 1 and $T(n) = \Theta(n)$
- (●) Apply case 2 and $T(n) = \Theta(n \log n)$
- ( ) Apply case 3 and $T(n) = \Theta(n)$
- ( ) Master method cannot be used.

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

QUESTION 13

The asymptotic relationship between $T_1(n) = 4T(n/2) + n^2$ and $T_2(n) = T(n/2) + 1$

- $T_1(n) = O(T_2(n))$
- **$T_1(n) = \Omega(T_2(n))$**
- $T_1(n) = \theta(T_2(n))$
- All the above are correct

QUESTION 14

The asymptotic relationship between $f(n) = 2^n$ and $g(n) = n^{10000}$

- $f(n) = O(g(n))$
- **$f(n) = \Omega(g(n))$**
- $f(n) = \theta(g(n))$
- All the above are correct

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

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QUESTION 15

1 pc

Using dynamic programming for **0/1 knapsack** problem, the time to choose which items that must be included in the knapsack is:

- O(Wn)
- O(n)
- O(W)
- All the above answers are incorrect.

QUESTION 16

1 pc

Finding the longest path between two point in the graph is:

- Optimization problem
- Decision problem
- Both answers a and b are correct
- Both answers a and b are incorrect

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Save All Answers

QUESTION 17

1 points

Using the **brute-force approach** to solve the Satisfiability (SAT) problem with 5 variables, we need to check _____ possible combinations of these items.

- 5
- 10
- 25
- 32 (selected)

QUESTION 18

1 points

The brute force algorithms:

- try all possible solutions.
- grantee to find the optimal (best) solutions
- are expensive
- All the above answers are correct (selected)

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QUESTION 19

The solution of any greedy algorithm is always optimal.

- ○ True
- ◉ False

QUESTION 20

Bellman-Ford algorithm is an example of greedy algorithm.

- ◉ True
- ○ False

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

QUESTION 21

Trying all the possible solutions of a problem except those that cannot be a valid solution is _____ algorithmic strategy.

- Dynamic programming
- Divide-and-conquer
- Backtracking
- Greedy

QUESTION 22

To prove that a problem X is an NP-complete, you need to prove that:

- $X \in NP$
- For all $Y \in NP, Y \leqslant_p X$
- Both answer **a** and **b**
- All the above answers are incorrect

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All the above answers are incorrect

QUESTION 23

Problems that can be solved in polynomial time are known as?

- intractable
- tractable
- decision
- NP-complete

QUESTION 24

If any NP-complete problem can be solved in polynomial time, then every problem in NP can also be solved similarly.

- True
- False

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

True

False

QUESTION 25

Before designing an algorithm, you should decide first whether the solution is exact or approximate.

- True
- False

QUESTION 26

To solve the 3-SAT problem in reasonable time, you should not use:

- an approximation algorithm
- heuristic approach
- Brute force approach (check every combination of the variables)
- All the answers are correct

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Save All

- an approximation algorithm
- heuristic approach
- (●) Brute force approach (check every combination of the variables)
- All the answers are correct

QUESTION 27

It has been proven that P=NP.

- True
- (●) False

QUESTION 28

Let X be a problem that belongs to the class NP. If X is not NP-hard, then it is not NP-complete.

- (●) True
- False

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

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Remaining Time: 02 minutes, 12 seconds.

Question Completion Status:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
27 28 29 30 31 32 33 34 35 36 37 38 39 40

QUESTION 29

Let a problem X is an NP problem and a problem Y is an NP-complete problem. If $Y \leqslant_p X$, then X is:

- NP-complete problem
- NP-hard problem
- (•) Both answer **a** and **b**
- All the above answers are incorrect

QUESTION 30

If you have a problem that is NP-complete, it is preferable to:

- search for an optimal solution to this question.
- (•) find a solution using approximate algorithms.
- search for a polynomial-time algorithm to solve this problem.
- All the above answers are incorrect

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Save All Answ

Remaining Time: 02 minutes, 09 seconds.

Question Completion Status:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
26 27 28 29 30 31 32 33 34 35 36 37 38 39 40

All the above answers are incorrect

QUESTION 31 1 points

Consider problems X and Y. If X is an NP problem and a polynomial time reducible to Y, then Y is an NP problem.

- True
- False

QUESTION 32 1 points

Showing that TSP $\leq_p$ HAM-CYCLE can be used in proving that TSP is an NP-compete problem.

- True
- False

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Save All Answers

Remaining Time: 02 minutes, 06 seconds.

Question Completion Status:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40

QUESTION 33

1 points

_____ algorithm is used to find the shortest paths in a graph that contains a negative weight edge that does not create negative-weight cycles.

- The Dijkstra
- Bellman-Ford
- Both the above answers are correct
- Both the above answers are incorrect

QUESTION 34

1 points

Applying Bellman-Ford algorithm to a graph with 10 vertices requires ____ passes.

- 4
- 9
- 10
- Cannot be determined

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Save All Answers | Save and S

Remaining Time: 02 minutes, 04 seconds.

Question Completion Status:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40

QUESTION 35

1 points

Select the correct statement about BFS and DFS algorithms:

- Time complexity of DFS is lower while the space complexity is the same.
- Time complexity of BFS is lower while the space complexity is the same
- Space complexity of DFS is lower while the time complexity is the same.
- Space complexity of BFS is lower while the time complexity is the same

QUESTION 36

1 points

The algorithm that is used to find the strongly connected component in a graph is

- Breadth first search.
- Depth first search.
- Prim
- Kruskal

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Save All Answers

Save and Sub

Remaining Time: 02 minutes, 02 seconds.

Question Completion Status:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40

Kruskal

QUESTION 37

1 points

Consider a **fractional knapsack problem** with four items:
Item 1: weight = 10 and benefit = 30
Item 2: weight = 5 and benefit = 25
Item 3: weight = 20 and benefit = 40
Item 4: weight = 5 and benefit = 20
The knapsack capacity = 50.
What isthe most efficient way to solve this problem?

- Brute force approach
- Dynamic programming
- Greedy algorithm
- All the above answers are incorrect

QUESTION 38

1 points

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Save All Answers | Save and Submit

Question Completion Status:

All the above answers are incorrect

QUESTION 38

1 points

Consider a **fractional knapsack problem** with four items:
Item 1: weight = 10 and benefit = 30
Item 2: weight = 5 and benefit = 25
Item 3: weight = 20 and benefit = 40
Item 4: weight = 5 and benefit = 20
The knapsack capacity = 50.
What is the total (maximum) benefit of the items included in the knapsack?

- 75
- 45
- 115
- 25

QUESTION 39

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Save All Answers | Save and Submit

Consider a **fractional knapsack problem** with four items:
Item 1: weight = 10 and benefit = 30
Item 2: weight = 5 and benefit = 25
Item 3: weight = 20 and benefit = 40
Item 4: weight = 5 and benefit = 20
The knapsack capacity = 50.
which items you will choose?

- Items 2 and 4 and part of item 1
- Only item 2
- Items 1, 2, and 4 and part of item 3
- All the four items are included in the knapsack

QUESTION 40

1 points Saved

Consider a **fractional knapsack problem** with four items:
Item 1: weight = 10 and benefit = 30
Item 2: weight = 5 and benefit = 25
Item 3: weight = 20 and benefit = 40
Item 4: weight = 5 and benefit = 20
The knapsack capacity = 50.

QUESTION 40

1 points Saved

Consider a **fractional knapsack problem** with four items:
Item 1: weight = 10 and benefit = 30
Item 2: weight = 5 and benefit = 25
Item 3: weight = 20 and benefit = 40
Item 4: weight = 5 and benefit = 20
The knapsack capacity = 50.
What is the total weight of items included in the knapsack?

- 5
- 20
- 40
- 50

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Save All Answers | Save and Submit

Consider problems X and Y. If X is an NP problem and a polynomial time reducible to Y, then Y is an NP problem.

- (●) True
- ( ) False

QUESTION 32

Showing that TSP $\leqslant_p$ HAM-CYCLE can be used in proving that TSP is an NP-compete problem.

- ( ) True
- (●) False

QUESTION 33

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

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